Kubernetes Deployment Not Updating SOP

Objective

Ensure a Kubernetes deployment successfully rolls out new application versions when the expected image or manifest change is not reflected in pods.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T009
Issue	Kubernetes deployment not updating
Category	Deployment
Service	Kubernetes
Severity	Medium
Status	Resolved
Description	New image not reflecting in pods
Expected Resolution	Check image pull policy and rollout status

Procedure

1. Confirm the deployment name, namespace, desired image tag, and release timestamp.
2. Check rollout status, replica status, and deployment events.
3. Inspect the deployment image field and compare it with the expected build artifact.
4. Review imagePullPolicy, image tag reuse, and image registry availability.
5. Check ReplicaSets to confirm whether a new ReplicaSet was created.
6. Force a rollout restart or update the image tag when image caching prevents the change.
7. Validate new pods are created, running, and serving expected version.
8. Update ticket with rollout revision and final deployed image.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Deployment not updating	Image tag reused	Use unique immutable image tag
Pods still on old image	imagePullPolicy issue	Set proper pull policy or change tag
Rollout stuck	New pods failing readiness	Check pod events and app health
No new ReplicaSet	Manifest not applied	Reapply manifest or fix pipeline

Best Practices

1. Use immutable image tags for production releases.
2. Capture rollout revision in deployment logs.
3. Run post-deployment smoke tests.
4. Keep rollback command ready for failed rollouts.

References

Kubernetes Docs | Internal Tickets CSV | Cluster Events | Monitoring Dashboards | Application Logs | Platform Runbook